

Vignesh Edithal

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[in](https://www.linkedin.com/in/edithal) [linkedin.com/in/edithal](https://www.linkedin.com/in/edithal) [◇](https://github.com/edithal-14) github.com/edithal-14 [◇](https://github.com/edithal-14) [edithal-14.github.io](https://github.com/edithal-14)

WORK EXPERIENCE

Advanced Micro Devices (AMD)

November 2024 - present

Member of Technical Staff - ROCm Dev Tools - AI GPU Group

Markham, Ontario

- Built counter collection service interfacing with [rocprowler-sdk](#) to intercept kernel dispatches
- Implemented iteration multiplexing in [rocprowler-compute](#) with data imputation to circumvent hardware limitations
- Implemented PC sampling profiling with no app replay, enabling faster runs with ISA stall and hotspot analysis.
- Enabled profiling support for next-gen MI 350 GPUs by analyzing architecture and deriving performance metrics
- Designed SQL analysis database schema to organize derived metrics and enable baseline performance comparisons
- Automated environment setup using containers and standalone binary generation using Nuitka compiler

Advanced Micro Devices (AMD)

May 2023 - November 2024

Senior Software Engineer - Radeon Software Team

Markham, Ontario

- Finetuned CLIP and Auto Encoder models to detect interesting events in FPS gameplay with > 90% accuracy
- Trained EfficientNet models to detect visual quality artifacts in video games with > 95% correlation
- Designed efficient data collection methodology to develop > 50 GB in-house video dataset in a month
- Optimized state-of-the-art Transformer and CNN models for deployment using ONNX framework
- Improved chatbot experience using GraphRAG for efficient document search using Langchain API
- Developed automated scripts to collect 100s of GB of gameplay data using Python based OCR scripts and OBS Studio
- Visual Quality Assessment pipeline using EfficientNet distillation inspired from [Google's UVQ model](#) and [Netflix's VMAF scores](#)

D. E. Shaw

July 2018 - July 2022

Senior Member Technical

Hyderabad, India

- Enabled firm-wide hierarchical permission management feature using Python and SQL backend
- Microservices using Kafka, orchestration using Kubernetes and Docker, telemetry using Prometheus and Grafana
- Automated third party file transfers for HR using Python/Bash scripting leading to 10% ticket reduction
- Provided SRE support for production Linux RHEL cluster by debugging networking, storage, services issues

EDUCATION

University of Toronto

September 2022 - December 2023

MSc in Applied Computing: Deep Learning, Computer Vision, Data Science **GPA: 4.0/4.0**

IIT Madras

December 2021 - August 2022

Diploma in Data Science: Statistics, Machine Learning, Business Analytics **GPA: 9.3/10**

IIT Patna

September 2014 - May 2018

BTech in Computer Science: Algorithms, Databases, Operating Systems **GPA: 8.9/10**

PROJECT WORK

Brain Tumor MRI synthesis using aggregation of GANs [[GitHub](#)] [[Report](#)] **University of Toronto**

- Implemented and trained DCGAN and WGAN models from scratch with tuned hyper-parameters
- Pre-processed data using up/down-sampling and used style transfer for post processing
- Obtained PSNR and SSIM scores comparable to relevant literature
- Showcase research in poster presentation with industry/faculty and submitted conference format report

Machine learning to predict college drop out rate [[GitHub](#)]

IIT Madras, India

- Exploratory data analysis followed by data cleaning and transformation using scaling pipelines
- Evaluation and Hyper-parameter tuning of 10 ML algorithms from scikit-learn library
- Identify best performing combination of algorithm and parameter set for the classification problem

- Grid search tuned XGBoost regression to independently model frequency and severity of insurance policy
- Geographical clustering using DBSCAN. Data Imputation using Gaussian Mixture model. Frequency encoding
- Model validation using synthetic data from SDV library Copula. Tune profit loading in a competitive setting

PUBLICATION AND RESEARCH

Novelty Goes Deep. A Deep Neural Solution To Document Level Novelty Detection. (Ghosal, Edithal et. al., COLING 2018). ([aclanthology.org/C18-1237](#))

Is your document novel? Let attention guide you. An attention-based model for document-level novelty detection. (Ghosal, Edithal et. al., Natural Language Engineering 2022). ([Journal article](#))

Semantically-Aware Game Image Quality Assessment. (Zhu, Edithal et. al.). [arxiv paper](#)

Gameplay Highlights Generation (Edithal et. al.). [arxiv paper](#)

TECHNICAL STRENGTHS

GPU Performance Profiling	GPU Micro-architecture, Roofline, Occupancy analysis, Bandwidth analysis
AI and Deep learning	Transformers, PyTorch, Keras, Hugging Face, ONNX, Langchain
Software Engineering	Docker, Kubernetes, Kafka, PyTest, Copilot, JIRA, Git
Computer Vision	OpenCV, TorchVision, Decord, Scikit-image
Data Science	Pandas, Numpy, Matplotlib, Scikit-learn, SQL Alchemy
Devops	Github actions, Jenkins, Artifactory
Scripting	Bash, PowerShell, Python